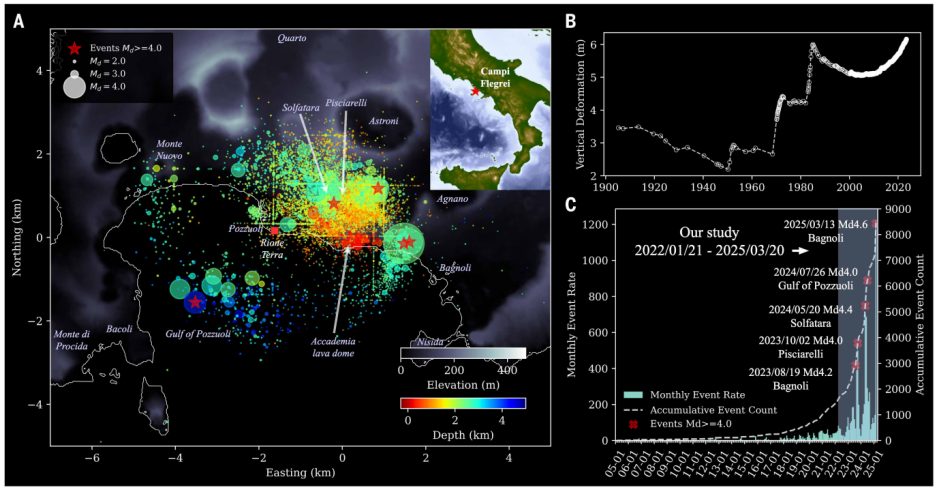
Figure 1: Seismicity and ground displacement

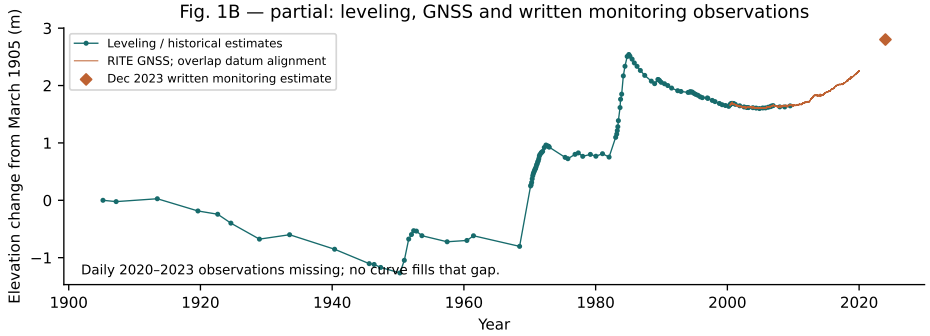
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

Panel 1B — partially reproduced



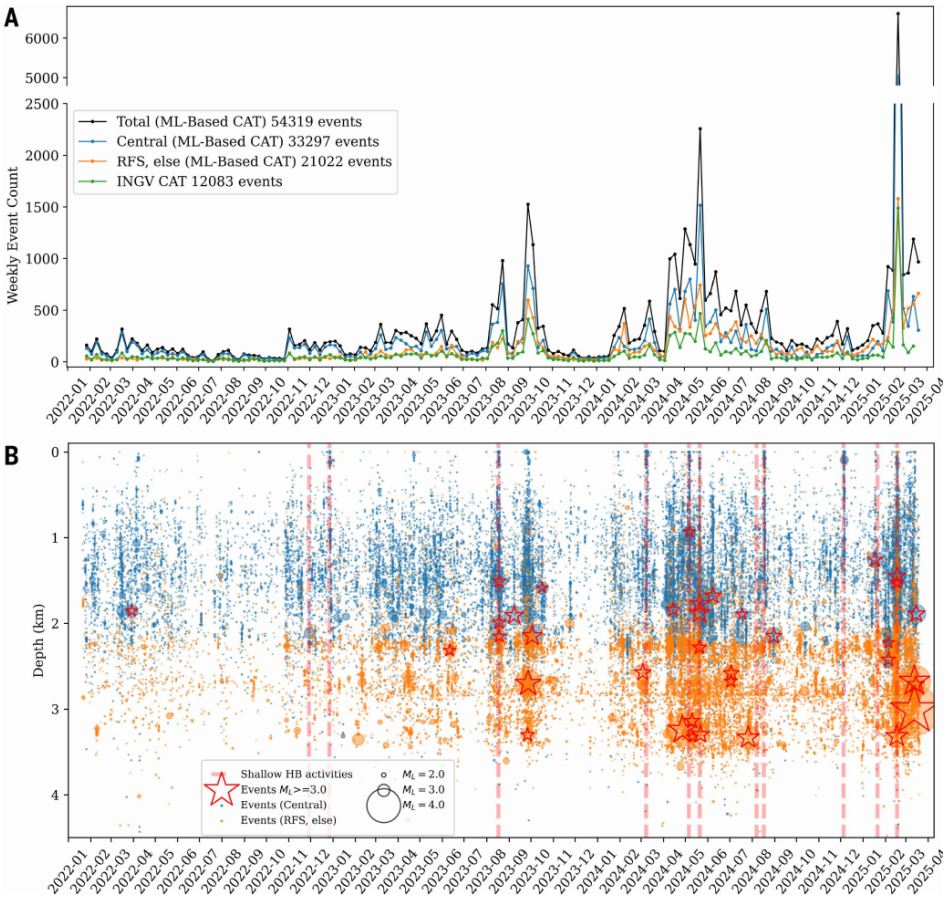
Panel 1B is partial: original leveling through 2009 and daily GNSS through 2019, with a separate written December 2023 monitoring endpoint. The vertical datum is relative to March 1905; daily observations for 2020–2023 are missing.

Panels A and C — not reproduced

Panels 1A and 1C have no valid independent counterpart; existing routine-catalog plots are external context.

Figure 2: Temporal evolution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 3.
DOI: 10.1126/science.adw9038.

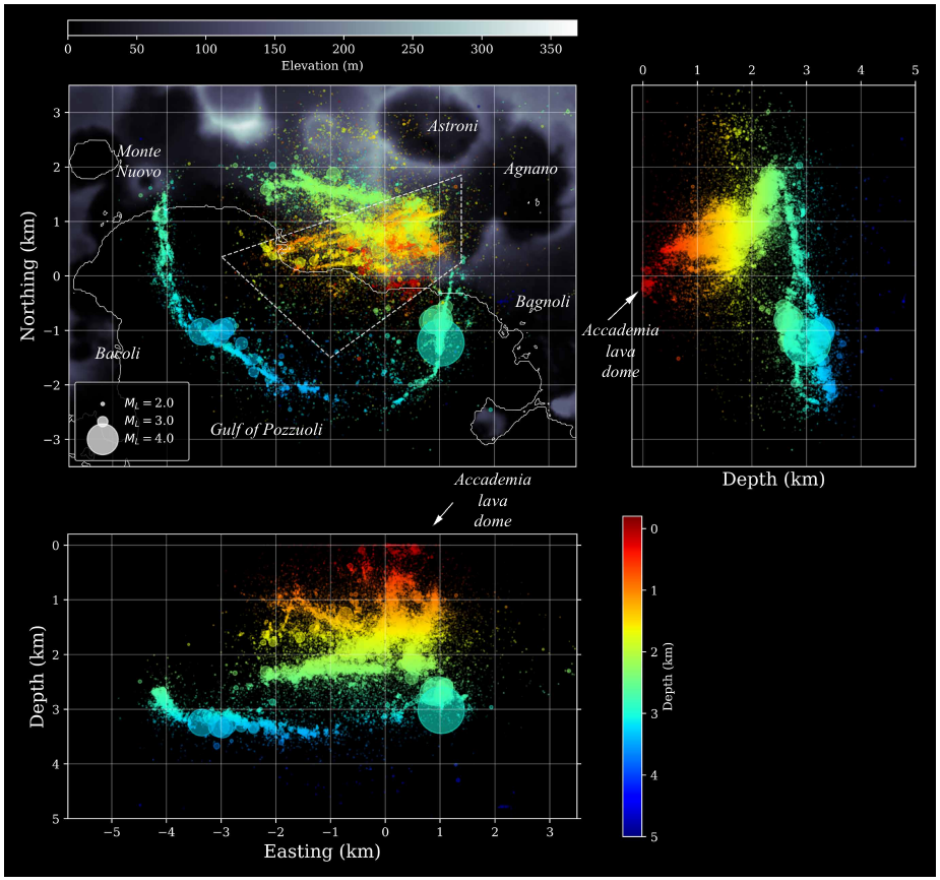
Independent result

All panels/views — not reproduced

Panels A and B require a multiyear calibrated, relocated catalog and magnitudes. Only nine provisional associations exist.

Figure 3: Spatial distribution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

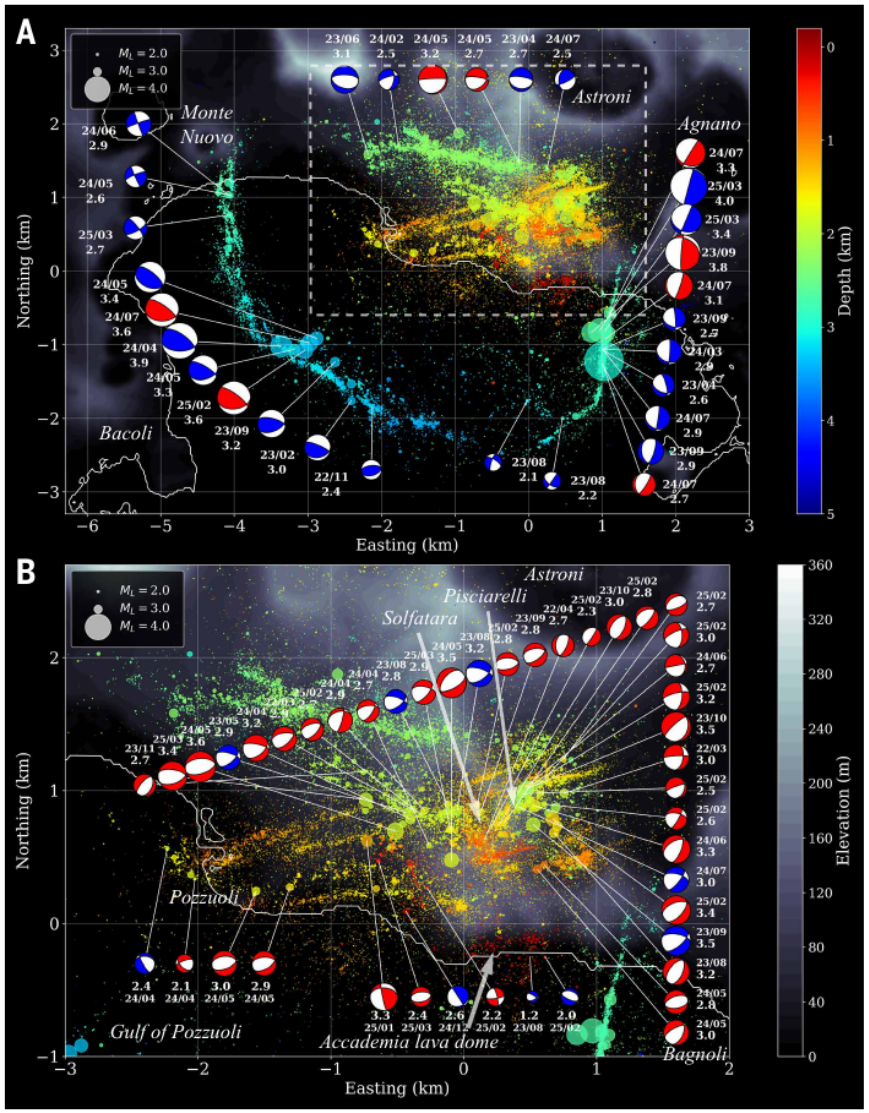
Independent result

All panels/views — not reproduced

The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 4: Focal mechanisms of larger events

Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

All panels/views — not reproduced

Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Figure S1: Recorded station availability

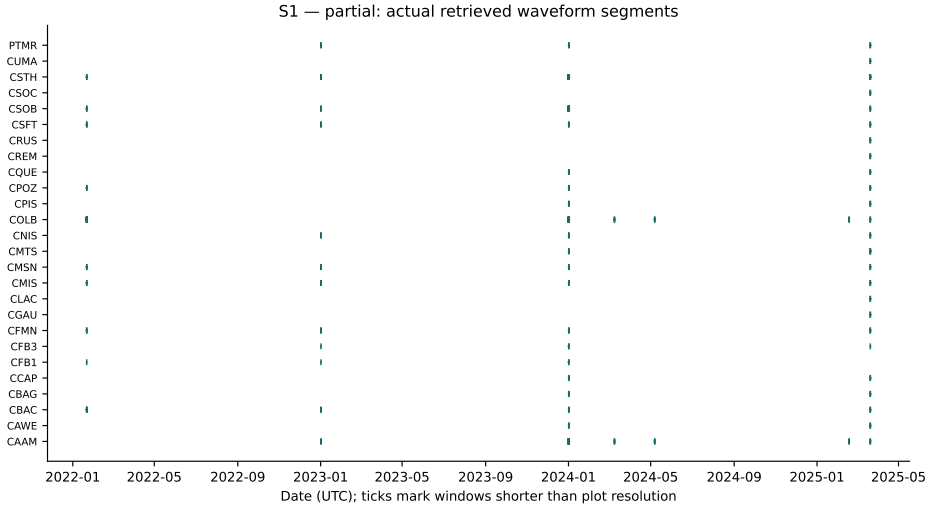
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

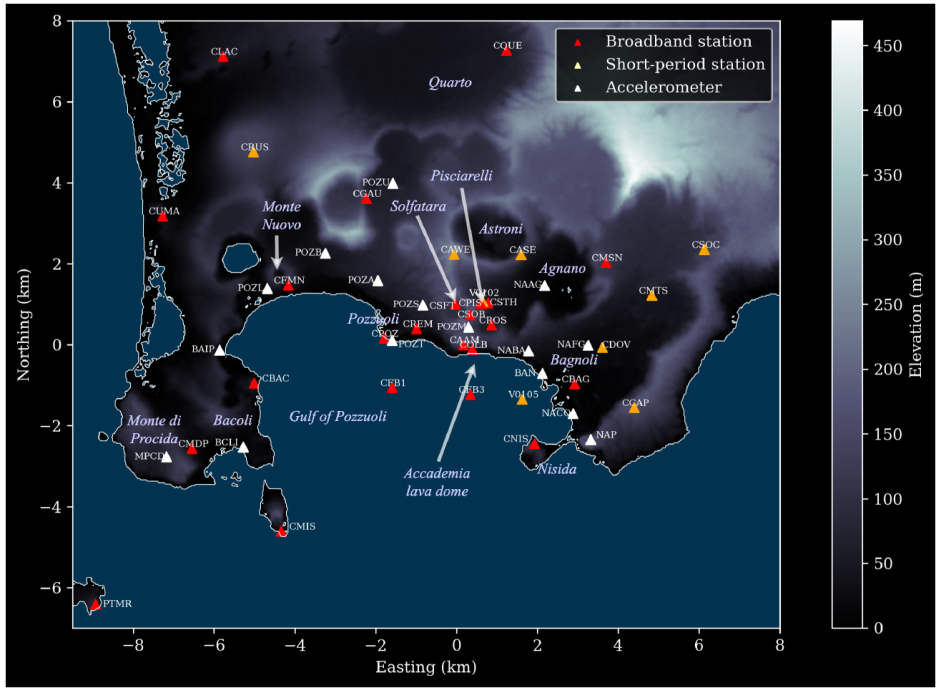
Panel S1 — partially reproduced



Partial: actual returned waveform intervals from 171 files and 26 station names. Short retrieved windows are marked without implying continuous coverage between them; channel-hours are not full network-years.

Figure S2: Station locations

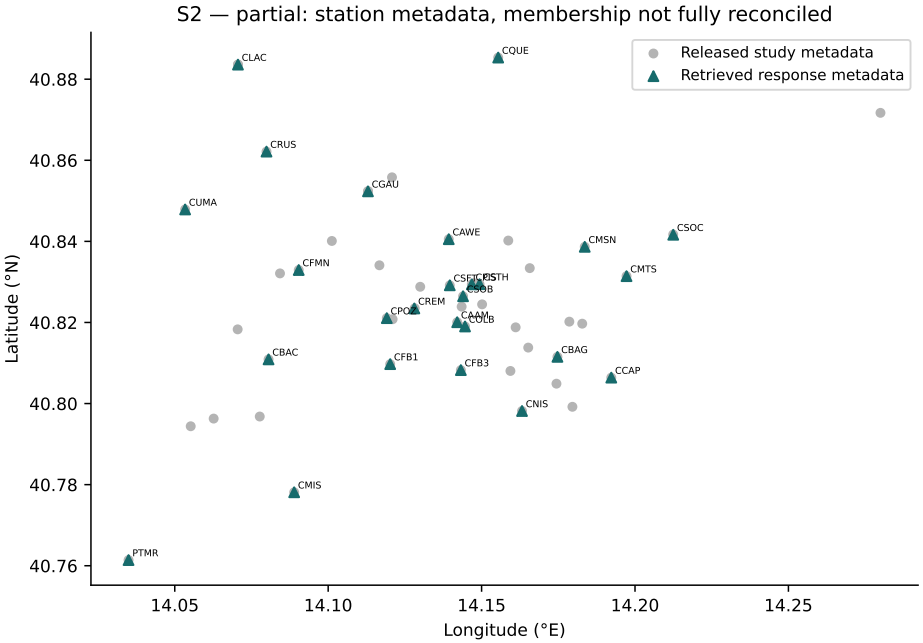
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

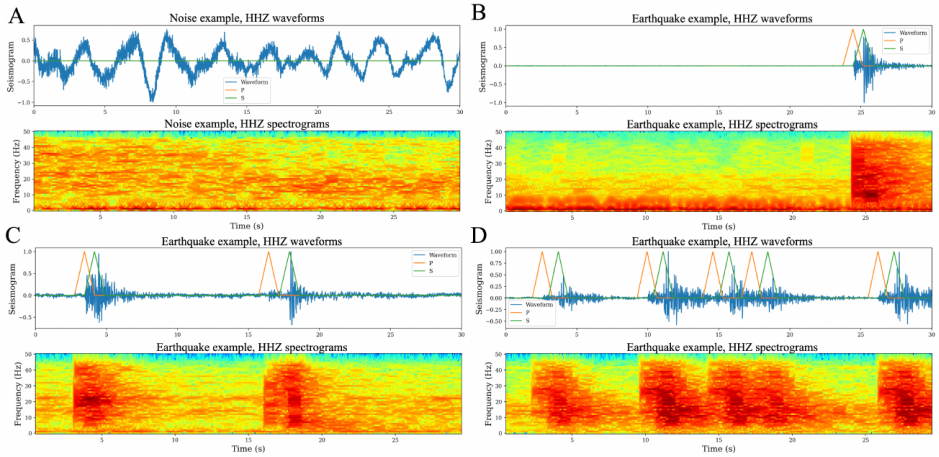
Panel S2 — partially reproduced



Partial: released station coordinates and retrieved IV/2I response metadata. Exact membership of the original 48 instruments, station epochs and the basemap are not fully reconciled.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

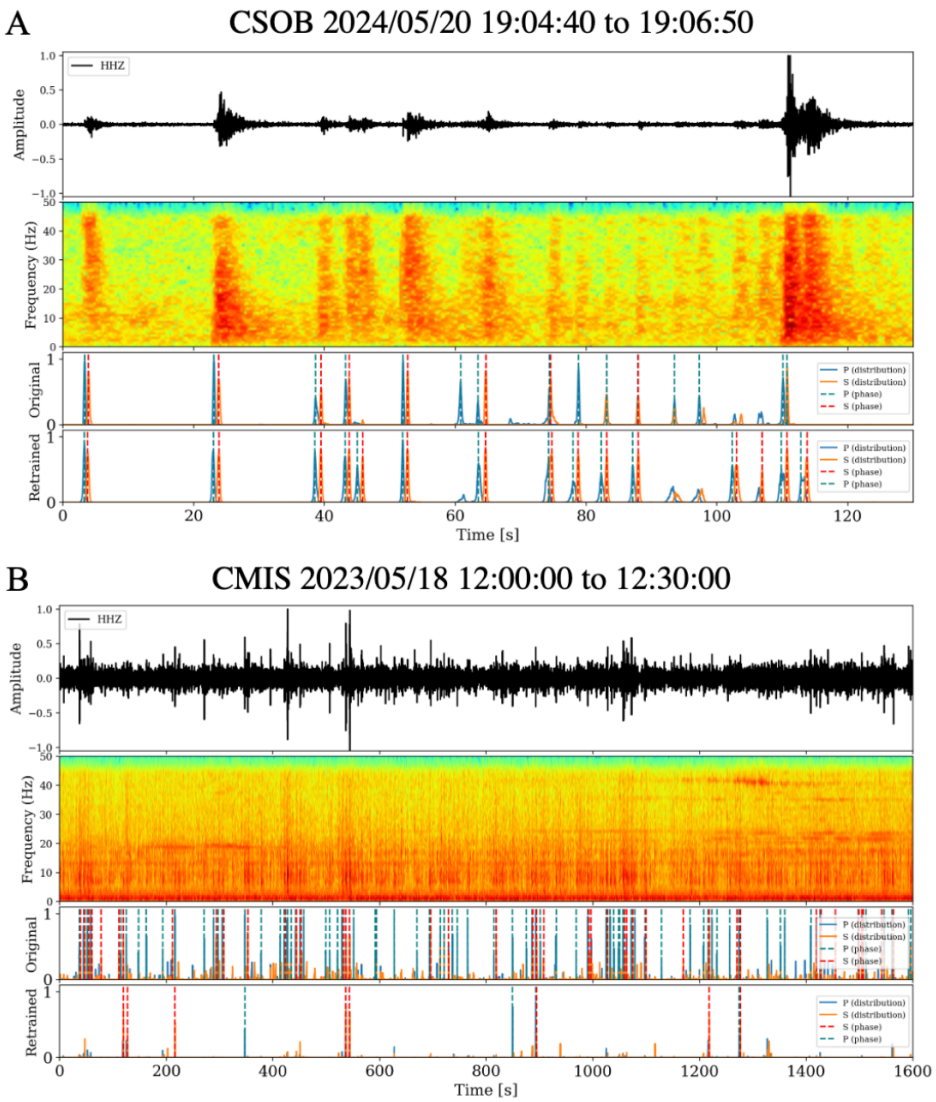
Independent result

All panels/views — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

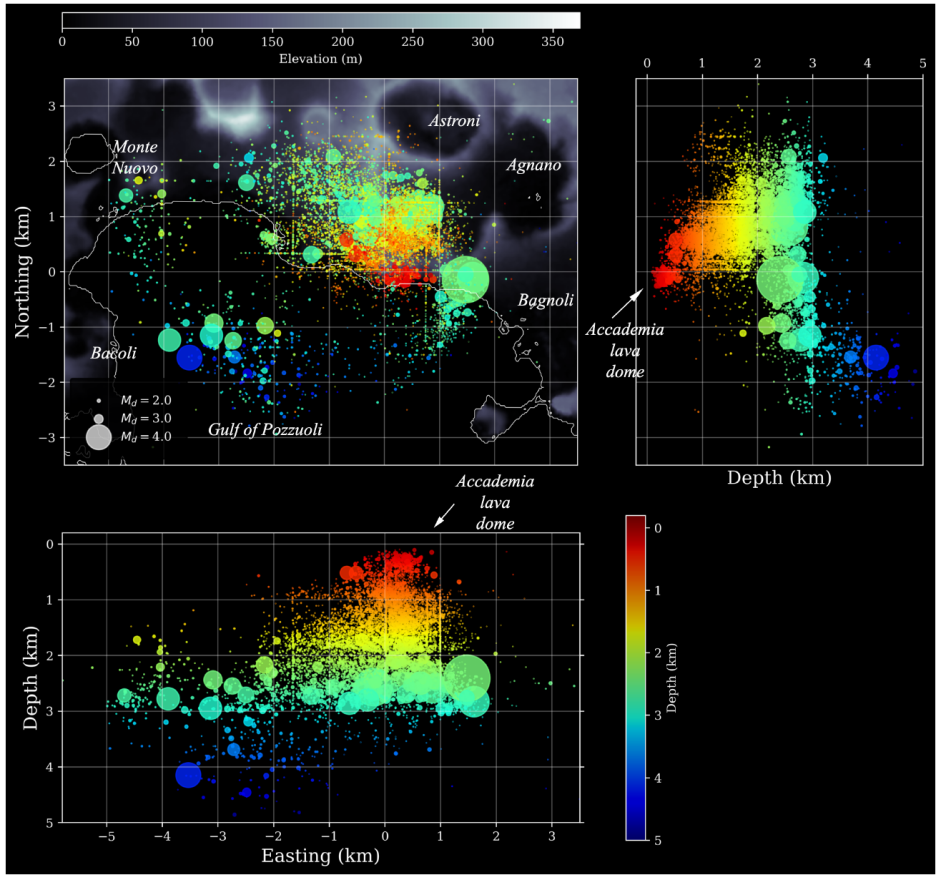
Independent result

All panels/views — not reproduced

Panels A and B were not generated for the exact CSOB swarm and CMIS noise examples. Other unlabeled inference comparisons remain diagnostics.

Figure S5: Routine-catalog locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

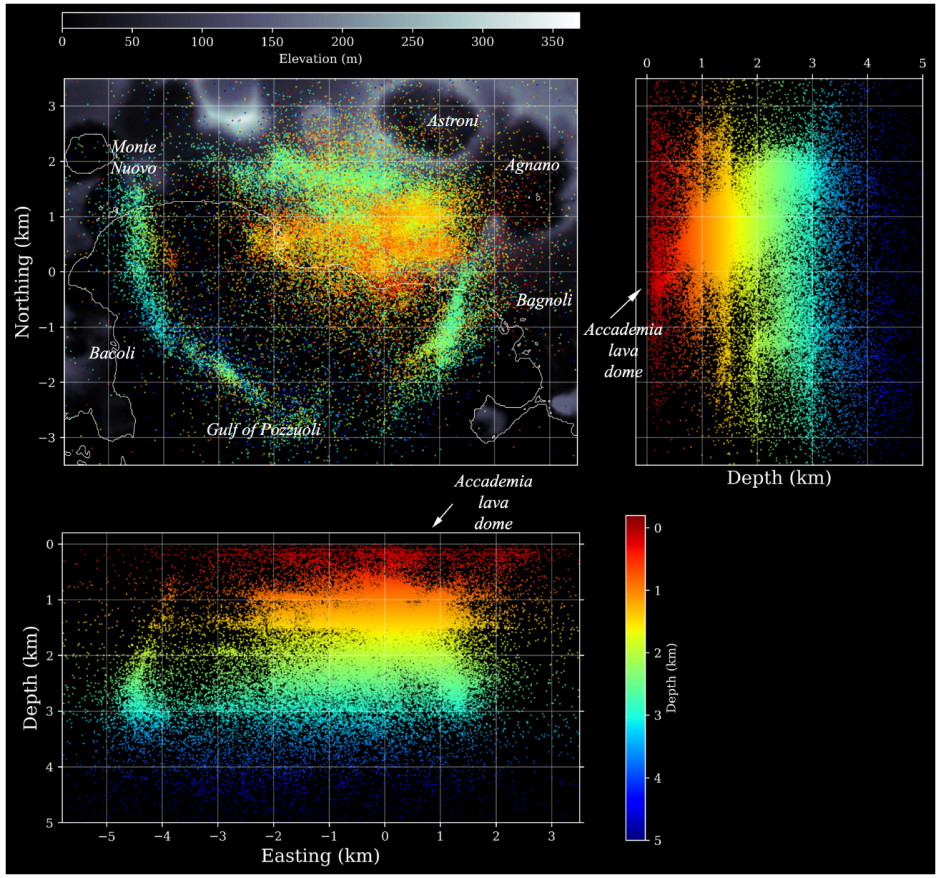
Independent result

All panels/views — not reproduced

Current INGV routine-catalog plots are external context. The saved panel record contains no valid independent counterpart to these three views; the published selection/version is unresolved.

Figure S6: Initial VELEST locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

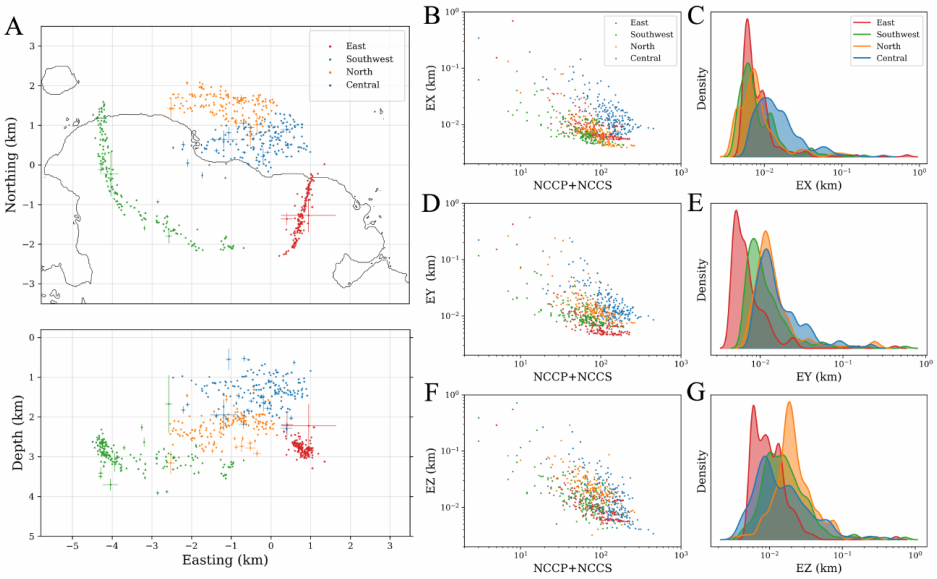
Independent result

All panels/views — not reproduced

The map and both cross sections require scientific VELEST location and velocity/station-correction runs. Compilation alone did not produce these results.

Figure S7: Location uncertainties

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

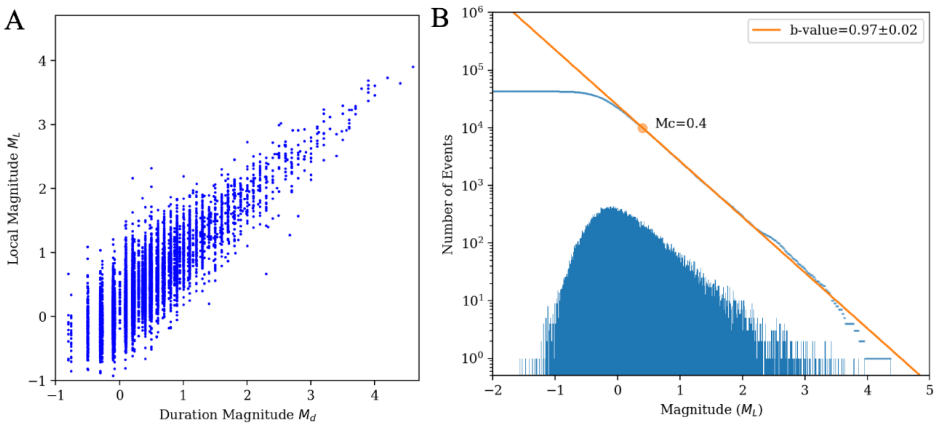
Independent result

All panels/views — not reproduced

Panels A–G require independently selected regional subsets, HypoDD SVD locations and coordinate uncertainties. None of these scientific inversions ran.

Figure S8: Magnitude calibration and frequency distribution

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

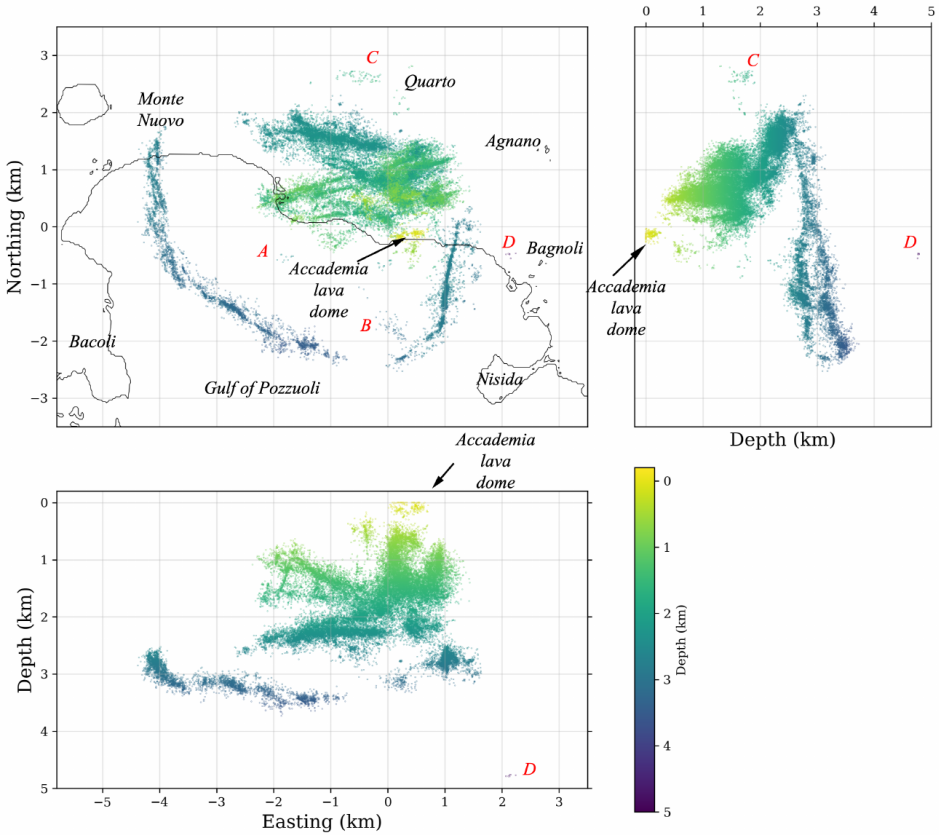
Independent result

All panels/views — not reproduced

Panels A and B require qualifying relative-amplitude equations and calibrated magnitudes. Zero qualifying equations were obtained; completeness and b-value were not estimated.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

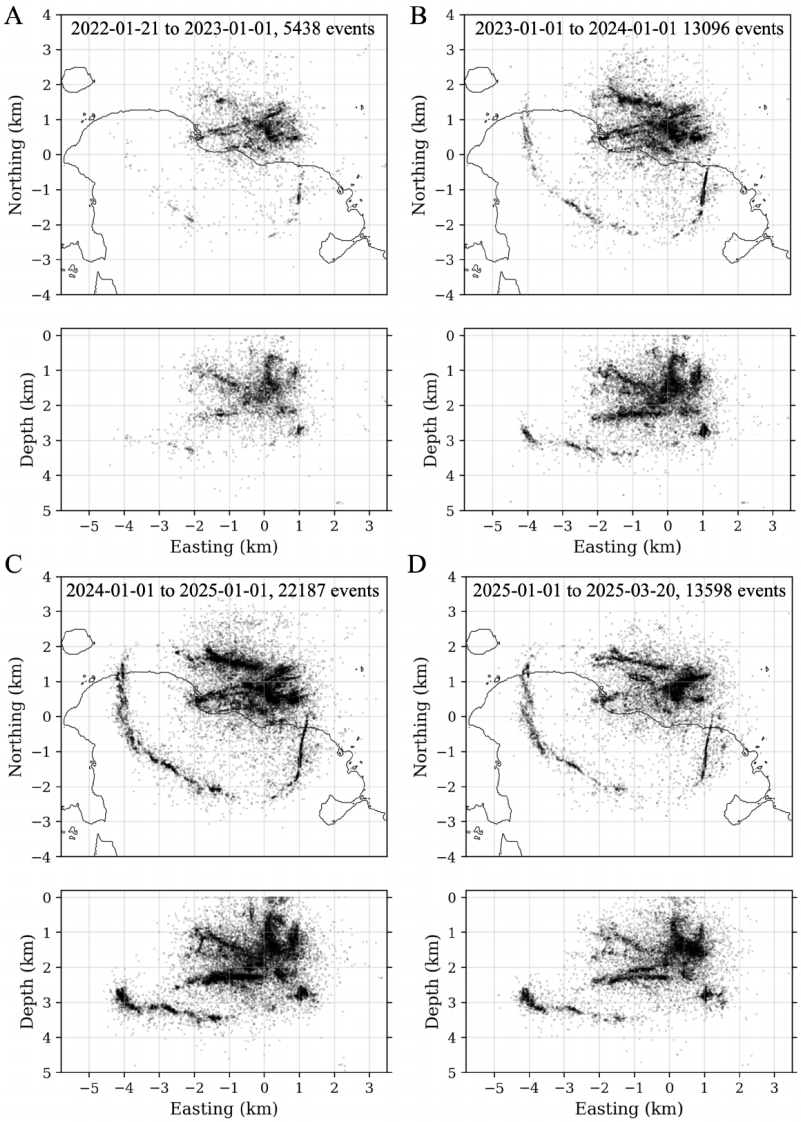
Independent result

All panels/views — not reproduced

All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

Figure S10: Seismicity by year

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

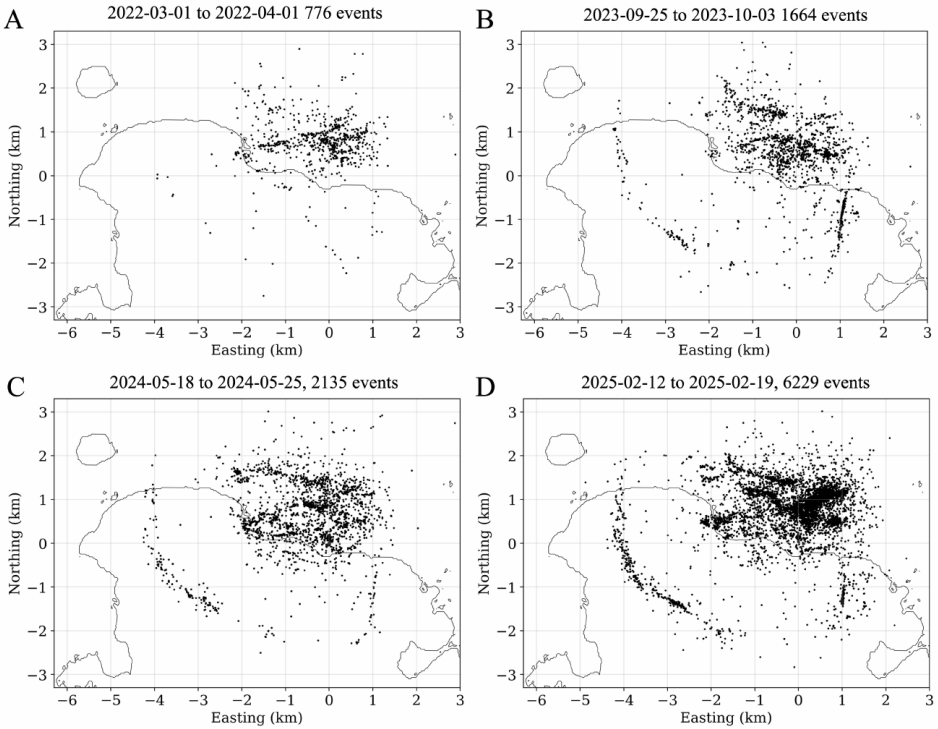
Independent result

All panels/views — not reproduced

The four annual groups for 2022–2025 require the independent multi-year relocated catalog. The available sample does not support them.

Figure S11: Four major swarms

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

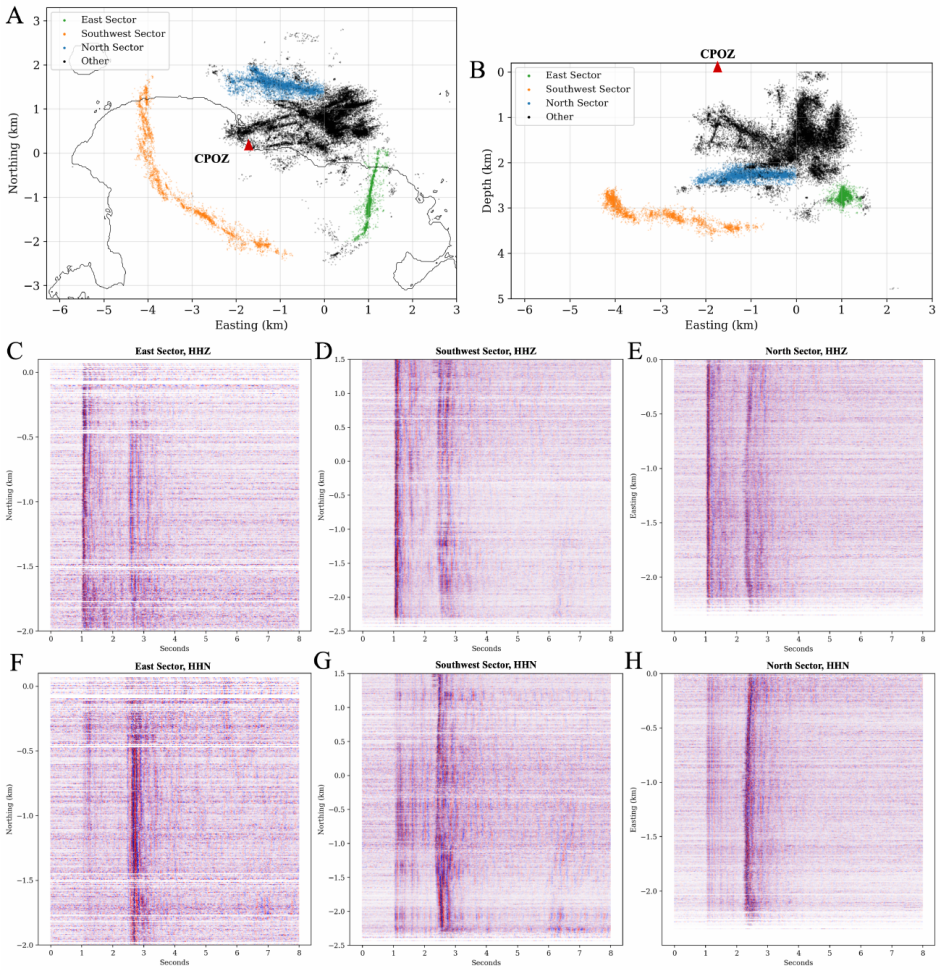
Independent result

All panels/views — not reproduced

The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S12: Ring sectors and CPOZ common-receiver gathers

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

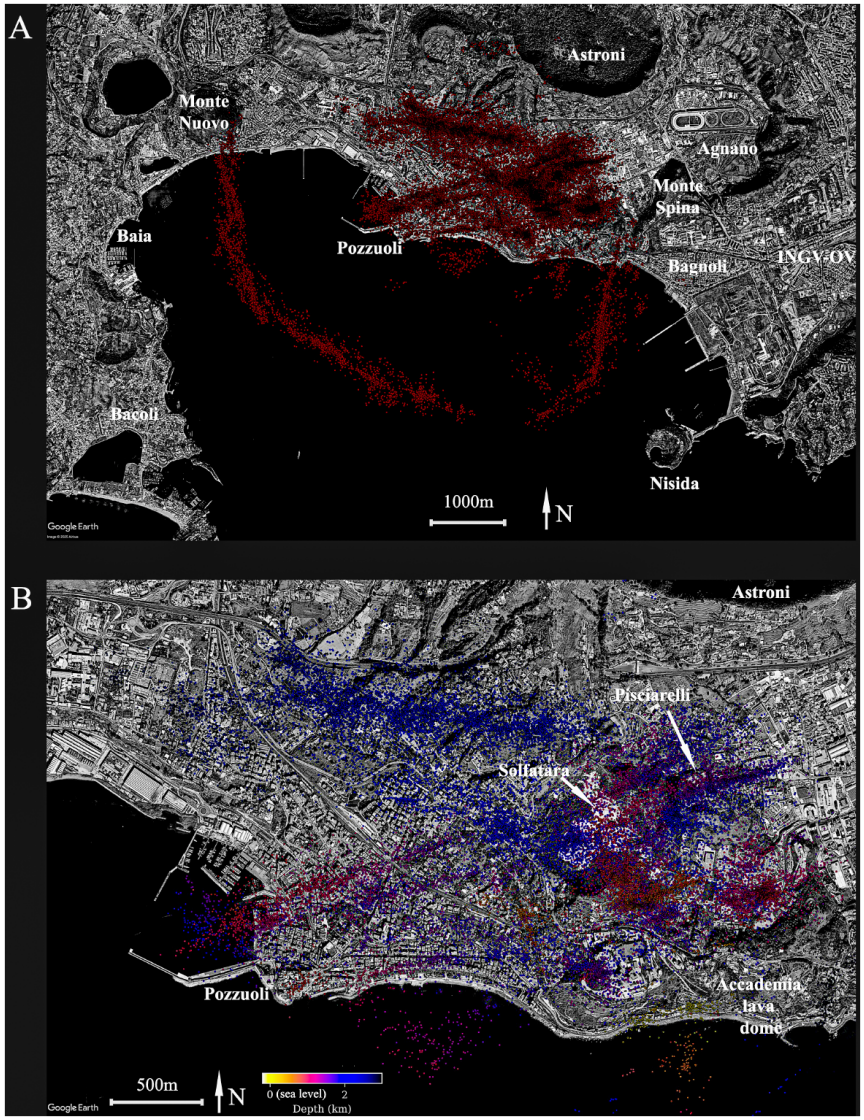
Independent result

All panels/views — not reproduced

Panels A–H require relocated sector selections and independently associated CPOZ P arrivals. The current associations contain zero CPOZ P arrivals.

Figure S13: Relocated seismicity on satellite imagery

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

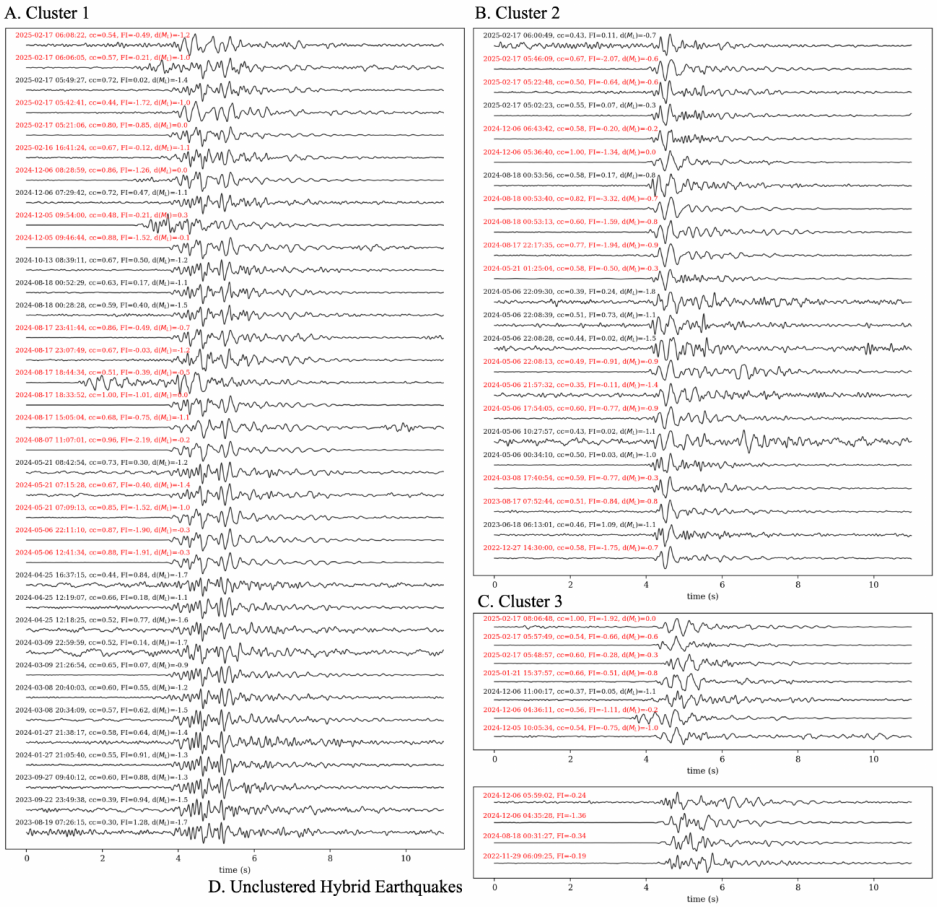
Independent result

All panels/views — not reproduced

Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

Figure S14: Hybrid-event clusters

Published reference



Independent result

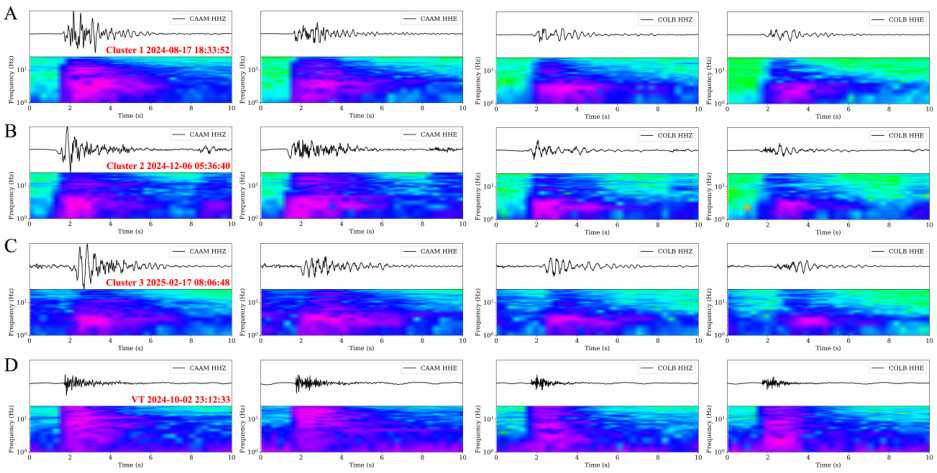
All panels/views — not reproduced

The three clusters and unclustered group require a multiyear scan, confirmed events, frequency indices and cluster-relative magnitudes. Four current candidates are unconfirmed.

Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

Figure S15: Hybrid and shallow-event waveforms

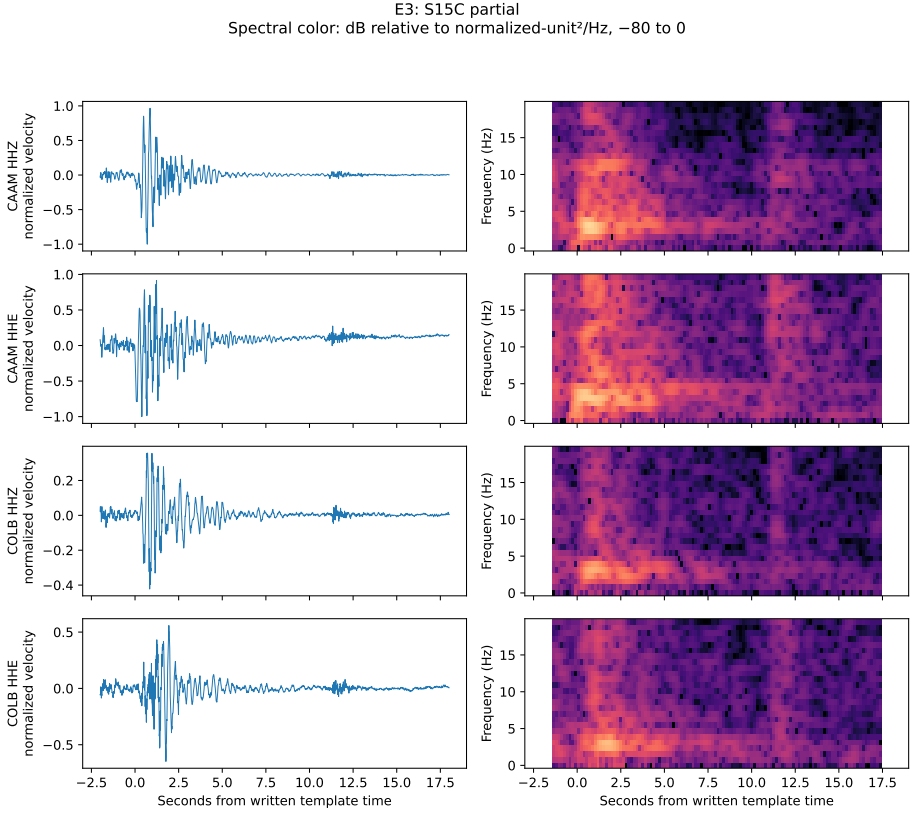
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

Panel S15C — partially reproduced



Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Panels A, B and D — not reproduced

Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.